Infinite Series 3

Siddharth K

January 2026

Problem 2

Rewriting $D_N(t)$ in terms of exponentials:

$$D_N(t) = \frac{1}{2} + \sum_{n=1}^{N} \cos(nt) = \frac{1}{2}\left(1 + \sum_{n=1}^{N}(e^{int} + e^{-nt})\right)$$

$$= \frac{1}{2}\sum_{n=-N}^{N} e^{int}.$$

This is a geometric series:

$$a = e^{-iNt}, \quad r = e^{it},$$

$$a\sum_{n=0}^{2N} r^n = \frac{a(r^{2N+1}-1)}{r-1} = \frac{e^{-iNt}(e^{(2N+1)t-1})}{e^{it}-1} = \frac{e^{i(N+1)t} - e^{iNt}}{e^{it}-1}.$$

Multiplying the numerator and denominator by $e^{-it/2}$:

$$= \frac{e^{i(N+1/2)t} - e^{i(N+1/2)t}}{e^{it/2} - e^{-it/2}} = \frac{2\sin((N+1/2)t)}{2\sin(t/2)}.$$

Finally, multiplying by $1/2$:

$$D_N(t) = \frac{\sin((N+1/2)t)}{2\sin(t/2)}.$$

Problem 3

$$\int_0^\pi D_N(t)dt = \int_0^\pi \left(\frac{1}{2} + \sum_{n=1}^{N}\cos(nt)\right)dt = \frac{\pi}{2} + \sum_{n=1}^{N}\int_0^\pi \cos(nt)dt$$

$$= \frac{\pi}{2} + \sum_{n=1}^{N} -\frac{1}{n}\sin(\pi n) = \frac{\pi}{2}.$$

Problem 5

Because f is even, the Fourier series has only cosine terms:

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx.$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos(ax)dx = \frac{1}{2\pi}\left[\frac{\sin(ax)}{a}\right]_{-\pi}^{\pi} = \frac{1}{2\pi}\frac{2\sin(a\pi)}{a\pi}.$$

To compute a_n, start with product-to-sum:

$$\cos(ax)\cos(nx) = \frac{1}{2}(\cos((a-n)x) + \cos((a+n)x)).$$

So

$$a_n = \int_{-\pi}^{\pi} \cos(ax)\cos(nx)dx = \frac{1}{2\pi}\int_{-\pi}^{\pi}(\cos((a-n)x) + \cos((a+n)x))dx$$

$$= \frac{1}{2\pi}\left(\frac{2\sin((a-n)\pi)}{a-n} + \frac{2\sin((a+n)\pi)}{a+n}\right) = \frac{1}{\pi}\left(\frac{\sin((a-n)\pi)}{a-n} + \frac{\sin((a+n)\pi)}{a+n}\right).$$

Now, simplifying the sines:

$$\sin((a\pm n)\pi) = \sin(a\pi \pm n\pi) = \sin(a\pi)\cos(a\pi) \pm \cos(a\pi)\sin(a\pi) = \sin(a\pi)(-1)^n,$$

$$a_n = \frac{1}{\pi}\left(\frac{\sin(a\pi)(-1)^n}{a-n} + \frac{\sin(a\pi)(-1)^n}{a+n}\right) = \frac{\sin(a\pi)(-1)^n}{\pi}\left(\frac{1}{a-n} + \frac{1}{a+n}\right).$$

$$= \frac{2a\sin(a\pi)(-1)^n}{\pi(a^2-n^2)}.$$

So the full Fourier series is

$$\cos(ax) = \frac{\sin(a\pi)}{a\pi} + \sum_{n=1}^{\infty}\left(\frac{2a\sin(a\pi)(-1)^n}{\pi(a^2-n^2)}\right)\cos(nx).$$

Since f is defined on $[-\pi, \pi)$ and extended periodically, we have that

$$f(\pi) = f(-\pi) = \cos(-a\pi) = \cos(a\pi).$$

Therefore, the Fourier series at π converges to $\cos(a\pi)$. At $x = \pi$, $\cos(n\pi) = (-1)^n$, so

$$\cos(a\pi) = \frac{\sin(a\pi)}{a\pi} + \sum_{n=1}^{\infty}\left(\frac{2a\sin(a\pi)(-1)^n}{\pi(a^2-n^2)}\right)(-1)^n$$

$$= \frac{\sin(a\pi)}{\pi}\left(\frac{1}{a} + \sum_{n=1}^{\infty}\frac{2a}{a^2-n^2}\right).$$

Since $a \notin \mathbb{Z}$, we can divide both sides by $\sin(a\pi)$:

$$\pi\cot(a\pi) = \frac{1}{a} + \sum_{n=1}^{\infty}\frac{2a}{a^2-n^2},$$

which is the cotangent sum identity.

Problem 9

When the period is p, the frequency becomes $\frac{2\pi}{p}$, so the exponent becomes $e^{i2\pi x/p}$. Given any starting point x_0, we integrate over one period:

$$c_n = \frac{1}{p}\int_{x_0}^{x_0+p} f(x)e^{-i2\pi nx/p}\,dx.$$

Similarly,

$$a_n = \frac{2}{p}\int_{x_0}^{x_0+p} f(x)\cos\left(\frac{2\pi nx}{p}\right)dx, \quad b_n = \frac{2}{p}\int_{x_0}^{x_0+p} f(x)\sin\left(\frac{2\pi nx}{p}\right)dx.$$

Problem 10

Part (b)

Let $u = \lambda t, dt = du/\lambda$:

$$\widehat{g}(\xi) = \int e^{-2\pi\xi t}f(\lambda t)dt = \frac{1}{\lambda}\int e^{-2\pi i(\xi/\lambda)u}f(u) = \frac{1}{\lambda}\widehat{f}(\xi/\lambda).$$

Part (c)

Take the Fourier transform:

$$\widehat{f*g}(\xi) = \int e^{-2\pi i\zeta t}\left(\int f(x)g(t-x)dx\right)dt$$

$$= \int f(x)\left(\int e^{-2\pi i\zeta t}g(t-x)dt\right)dx.$$

Substitute $u = t - x$:

$$= \int f(x)e^{-2\pi i\xi x}\left(\int e^{-2\pi\xi u}g(u)du\right)dx = \widehat{f*g}(\xi) = \widehat{f}(\xi)\widehat{g}(\xi).$$

Part (d)

$$\widehat{f}'(\xi) = \int e^{-2\pi i\zeta t}f'(t)dt = \left[e^{-2\pi i\xi t}f(t)\right]_{-\infty}^{\infty} + \int(2\pi i\xi)e^{-2\pi i\xi t}f(t)dt.$$

The boundary term vanishes because f decays faster than any power, so

$$\widehat{f}'(\xi) = \int(2\pi i\xi)e^{-2\pi i\xi t}f(t)dt = 2\pi i\xi\widehat{f}(\xi).$$

Problem 11

Since f is only nonzero on $(-\pi, \pi)$, the integral becomes

$$\widehat{f}(\xi) = \int_{-\pi}^{\pi} e^{-2\pi i \xi x} dx = \left[\frac{e^{-2\pi i \xi x}}{-2\pi i \xi} \right]_{-\pi}^{\pi} = \frac{e^{-2\pi i \xi \pi} - e^{2\pi i \xi \pi}}{-2\pi i \xi} = \frac{\sin(2\pi^2 \xi)}{\pi \xi}.$$

Problem 12

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} e^{-2\pi i \xi x} e^{-\pi x^2} dx = \int_{-\infty}^{\infty} e^{-2\pi i \xi x - \pi x^2} dx.$$

Completing the square:

$$-2\pi i \xi x - \pi x^2 = -\pi(x^2 + 2i\xi x),$$

$$x^2 + 2i\xi x = (x + i\xi)^2 - \xi^2,$$

$$-\pi(x^2 + 2i\xi x) = -\pi(x + i\xi)^2 + \pi \xi^2.$$

Therefore,

$$\widehat{f}(\xi) = e^{-\pi \xi^2} \int_{-\infty}^{\infty} e^{-\pi(x+i\xi)^2} dx = e^{-\pi \xi^2} \int_{-\infty}^{\infty} e^{-\pi y^2} dy$$

$$= e^{-\pi \xi^2} \int_{-\infty}^{\infty} e^{-(\sqrt{\pi} y)^2} dy = e^{-\pi \xi^2} \cdot \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2} du = e^{-\pi \xi^2}.$$

For $f_t(x) = e^{-\pi t x^2} = f(\sqrt{t} x)$, we can use the scaling property:

$$g(x) = f(\lambda x) \implies \widehat{g}(\xi) = \frac{1}{\lambda} \widehat{f}(\xi / \lambda).$$

We get

$$\widehat{f}_t(\xi) = \frac{1}{\sqrt{t}} e^{-\pi \xi^2 / t}.$$